

修士論文

修論のタイトル

自分の名前

2024.3

北海道大学大学院理学院
数学専攻博士前期課程

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Preface

In these notes, We study microlocal Morse theory, which is presented, for instance, in [GKS12]. However, we replace the notion of microsupports with that of truncated microsupprts, introduced by Kashiwara, Monteiro-Fernandes, and Schapira in [KMS03].

1 Notations

We consider a commutative unital ring $\mathbf{k}$ of finite global dimension (e.g. $\mathbf{Z}$ or a field). Let X be a real analytic manifold. We denote the cotangent bundle of X by $\pi: T^*X \rightarrow X$. We denote by $D^b(\mathbf{k}_X)$ the bounded derived category of sheaves of $\mathbf{k}$ -modules on X .

2 Truncated Microsupport

Definition 2.1. Let $F \in \mathbf{D}^b(\mathbf{k}_X)$. The closed conic subset $\mathrm{SS}_k(F)$ of T^*X is defined by: $p \notin \mathrm{SS}_k(F)$ if and only if there exists an open conic neighborhood U of p such that for any $x \in \pi(U)$ and for any real-valued C^1 -function φ defined on a neighborhood of x such that $\varphi(x) = 0$, $d\varphi(x) \in U$, one has

$$H_{\{\varphi \geq 0\}}^j(F)_x = 0 \quad \text{for any } j \leq k. \quad (2.1)$$

2.1 Functorial Properties

Proposition 2.2. Let $f: X \rightarrow Y$ be a morphism of real analytic manifolds and let $F \in \mathbf{D}^b(\mathbf{k}_X)$ such that f is proper on $\mathrm{supp}(F)$. Then for any $k \in \mathbf{Z}$,

$$\mathrm{SS}_k(\mathbf{R}f_*F) \subset f_{\pi}f_d^{-1}(\mathrm{SS}_k(F)). \quad (2.2)$$

The equality holds in case f is a closed embedding.

3 Lemmas from [KS90]

Consider a projective system of abelian groups $(M_n, \rho_{n,p})_n$ indexed by $\mathbf{N}$. One says that this system satisfies the Mittag-Leffler condition (ML for short) if for any $n \in \mathbf{N}$, the decreasing sequence $(\rho_{n,p}(M_p))_p$ of subgroups of M_n is stationary.

For a projective system of abelian groups $(M_n)_n$, we set

$$M_{\infty} := \varprojlim_n M_n.$$

Consider a projective system of complexes

$$E_n^{\bullet}: \dots \rightarrow M_n^{j-1} \rightarrow M_n^j \rightarrow M_n^{j+1} \rightarrow \dots, \quad (3.1)$$

and projective limit

$$E_{\infty}^{\bullet}: \dots \rightarrow M_{\infty}^{j-1} \rightarrow M_{\infty}^j \rightarrow M_{\infty}^{j+1} \rightarrow \dots. \quad (3.2)$$

Denote by

$$\Phi_k: H^k(E_{\infty}^{\bullet}) \rightarrow \varprojlim_n H^k(E_n^{\bullet}) \quad (3.3)$$

the natural morphism.

Proposition 3.1 ([KS90, Proposition 1.12.4]). Assume that for each $j \in \mathbf{Z}$, the system $(M_n^j)_n$ satisfies the ML condition. Then

- (a) for each $k \in \mathbf{Z}$, the map Φ_k is surjective,
- (b) if moreover, for a given i the system $(H^{i-1}(E_n^{\bullet}))_n$ satisfies the ML condition, then Φ_i is bijective.

The following is called Kashiwara's lemma.

Proposition 3.2 ([KS90, Proposition 1.12.6]). Let $(X_s, \rho_{s,t}: X_t \rightarrow X_s)_{(s, (s \leq t))}$ be a projective system of sets indexed by $\mathbf{R}$. Assume that for each $s \in \mathbf{R}$, the canonical maps

$$\lambda_s: X_s \rightarrow \varprojlim_{r < s} X_r$$

and

$$\mu_s: \varinjlim_{t > s} X_t \rightarrow X_s$$

are both injective (resp. surjective). Then all the maps ρ_{s_0, s_1} ($s_0 \leq s_1$) are injective (resp. surjective).

ML procedure for sheaves

Proposition 3.3 ([KS90, Proposition 2.7.1]). Let X be a topological space and let $F \in \mathbf{D}^+(\mathbf{Z}_X)$. Let $(U_n)_{n \in \mathbf{N}}$ be an increasing sequence of open subsets of X , $(Z_n)_{n \in \mathbf{N}}$ a decreasing sequence of closed subsets of X . Set $U := \bigcup_n U_n$, $Z := \bigcap_n Z_n$.

- (i) For any j , the natural map $\varphi_j: H_Z^j(U; F) \rightarrow \varprojlim_n H_{Z_n}^j(U_n; F)$ is surjective.
- (ii) Assume that for a given j , the projective system $(H_{Z_n}^{j-1}(U_n; F))_n$ satisfies the ML condition. Then φ_j is bijective.
- (iii) Let now $(X_n)_{n \in \mathbf{N}}$ be an increasing family of subsets of X satisfying $X = \bigcup_n X_n$ and $X_n \subset \text{Int}(X_{n+1})$ for all n . Assume that for a given j , the projective system $(H^{j-1}(X_n; F))_n$ satisfies the ML condition. Then the natural map $H^j(X; F) \rightarrow \varprojlim_n H^j(X_n; F)$ is bijective.

4 Results

In this section, let M be a manifold.^{*1}

Theorem 4.1. Let F in $\mathbf{D}^b(\mathbf{k}_M)$, and $\psi: M \rightarrow \mathbf{R}$ be a real-valued function of class C^1 , proper on $\text{supp}(F)$. Let $a < b$ be real numbers. We assume $d\psi(x) \notin \text{SS}_k(F)$ for $a \leq \psi(x) < b$. For $t \in \mathbf{R}$, we set $M_t := \psi^{-1}(]-\infty, t])$. Then, the restriction morphisms

$$H^j(M_b; F) \rightarrow H^j(M_a; F)$$

are isomorphic for each $j < k$.^{*2}

Lemma 4.2. Let $-\infty < a < b < +\infty$ be real numbers and $G \in \mathbf{D}^b(\mathbf{R})$. Let k be an integer. If

$$\left(H_{[t, +\infty[}^j(G) \right)_t \cong 0 \quad \text{for all } t \in [a, b[, \text{ and all } j \leq k.$$

Then one has

$$H^j(]-\infty, b[; G) \xrightarrow{\sim} H^j(]-\infty, a[; G)$$

for all $j < k$.^{*3}

^{*1}Can M be of class C^∞ ?

^{*2}For $j = k$, this morphism might be injective.

^{*3}For k , this morphism might be injective. (not written in this draft.)

Proof. Let $t \in [a, b[$. We will denote by I_t the open interval $] -\infty, t[$ and denote by Z_t the closed interval $] -\infty, t] (= \overline{I_t})$. By applying the functor $(\cdot)_{\overline{I_t}}$ to the distinguished triangle

$$\mathrm{R}\Gamma_{\mathbf{R}-I_t}(G) \rightarrow G \rightarrow \mathrm{R}\Gamma_{I_t}(G) \xrightarrow{+1} \quad \text{in } \mathrm{D}^b(\mathbf{k}_{\mathbf{R}}),$$

we have

$$(\mathrm{R}\Gamma_{[t, +\infty[}(G))_{\{t\}} \rightarrow G_{Z_t} \rightarrow \mathrm{R}\Gamma_{I_t}(G) \xrightarrow{+1} \quad \text{in } \mathrm{D}^b(\mathbf{k}_{\mathbf{R}}).$$

Applying the functor $\mathrm{R}\Gamma(\mathbf{R}; \cdot)$ to this, we have

$$(\mathrm{R}\Gamma_{[t, +\infty[}(G))_t \rightarrow \mathrm{R}\Gamma(Z_t; G) \rightarrow \mathrm{R}\Gamma(I_t; G) \xrightarrow{+1} \quad \text{in } \mathrm{D}^b(\mathbf{k}). \quad (4.1)$$

Then consider the following conditions for $j \in \mathbf{Z}$, $s \in [a, b[$, and $t \in]a, b[$:

$$\varinjlim_{t>s} H^j(I_t; G) \xrightarrow{\sim} H^j(I_s; G), \quad (a)_s^j$$

$$\varprojlim_{s<t} H^j(I_t; G) \xleftarrow{\sim} H^j(I_t; G). \quad (b)_t^j$$

By hypothesis, it follows from (4.1) that $H^j(Z_t; G) \xrightarrow{\sim} H^j(I_t; G)$ for $t \in [a, b[$.^{*4} Since $\varinjlim_{t>s} H^j(I_t; G) \cong H^j(Z_s; G)$, $(a)_s^j$ holds for any $j < k$ and $s \in [a, b[$.^{*5} Moreover, since we assume G is a bounded complex, $(b)_t^j$ holds for $j \ll 0$ and for all $t \in]a, b[$. Let us denote such j by j_0 . Let us argue by induction on j . Assume $(b)_t^j$ holds for all $t \in]a, b[$ and all $j \leq j_0$. Applying Proposition 3.2, we find the isomorphisms

$$H^j(I_s; G) \xrightarrow{\sim} H^j(I_t; G) \quad \text{for all } j \leq j_0 \text{ and all } a < s \leq t < b. \quad (4.2)$$

Let $G \xrightarrow{\sim} G^\bullet$ be a flabby resolution. Let $t \in]a, b[$. Consider a complex

$$E_n^\bullet = \Gamma(I_{t-\frac{1}{n}}; G^\bullet).$$

Since $G^\bullet$ is flabby, we have $E_n^j \cong E_\infty^j$ and the projective system $(E_n^j)_n$ satisfies the ML condition for all $j \in \mathbf{Z}$.

By (4.2), the projective system $(H^{j_0}(E_n^\bullet))_n$ satisfies the ML condition (because we have $a < t - \frac{1}{n} < t$ for $n \gg 0$). Applying Proposition 3.1 to $(E_n^\bullet)_n$, it follows that $(b)_t^{j_0}$ holds for $j_0 + 1$ and the induction proceeds. More precisely, we have

$$\begin{aligned} H^{j_0+1}(I_t; G^\bullet) &\xrightarrow{\sim} \varprojlim_{\Phi_{j_0+1}} \varinjlim_n H^{j_0+1}(I_{t-\frac{1}{n}}; G^\bullet) \\ &\cong \varprojlim_n H^{j_0+1}(I_{t-\frac{1}{n}}; G) \\ &\cong \varprojlim_{s<t} H^{j_0+1}(I_s; G). \end{aligned}$$

^{*4}Similarly, we have injections $H^k(Z_t; G) \hookrightarrow H^k(I_t; G)$ for $t \in [a, b[$.

^{*5}Similarly,

$$\varinjlim_{t>s} H^k(I_t; G) \hookrightarrow H^k(Z_s; G)$$

gives injections

$$\varinjlim_{t>s} H^k(I_t; G) \hookrightarrow H^k(I_s; G)$$

for all $s \in [a, b[$.

Again by Proposition 3.2, we get the isomorphisms

$$H^j(I_s; G) \xrightarrow{\sim} H^j(I_t; G) \quad \text{for all } j < k \text{ and all } a < s \leq t \leq b,$$

and injections

$$H^j(I_s; G) \hookrightarrow H^j(I_t; G)$$

for $j = k$ and for all $a < s \leq t \leq b$.

Finally, using (4.1) and the hypothesis, we have the isomorphisms

$$H^j(I_a; G) \xleftarrow{\sim} H^j(Z_a; G) \xrightarrow{\sim} H^j(I_s; G) \quad \text{for all } j < k \text{ and all } a < s \leq b.$$

Similarly, for $j = k$, we have

$$H^k(I_b; G) \cong \varinjlim_{t > s} H^k(I_t; G) \hookrightarrow H^k(I_s; G) \hookrightarrow H^k(I_a; G).$$

□

Proof of Theorem 4.1. First, we have

$$H^j(M_t; F) \cong H^j \mathrm{R}\Gamma(\cdot, t[; \mathrm{R}\psi_* F)$$

for $t \in I$. Since we assume that $d\psi(x) \notin \mathrm{SS}_k(F)$ for $a \leq \psi(x) < b$, and by (2.2), the formula $\mathrm{SS}_k(\mathrm{R}\psi_* F) \subset \psi_\pi \psi_d^{-1} \mathrm{SS}_k(F)$ holds. Hence we have

$$(H^j \mathrm{R}\Gamma_{[\psi(x), +\infty[}(\mathrm{R}\psi_* F))_{\psi(x)} = 0$$

for each $j \leq k$. Then we can apply Lemma 4.2 and have

$$H^j(\cdot, -\infty, b[; \mathrm{R}\psi_* F) \xrightarrow{\sim} H^j(\cdot, -\infty, a[; \mathrm{R}\psi_* F)$$

for all $j < k$. Then we can conclude that

$$H^j(M_b; F) \xrightarrow{\sim} H^j(M_a; F)$$

for all $j < k$.

□

5 Truncated Noncharacteristic Deformation

We will reformulate the noncharacteristic deformation lemma [KS90, Proposition 2.7.2] to make a relation to truncated microsupport.

Proposition 5.1. Let X be a Hausdorff space, $F \in \mathrm{D}^b(\mathbf{Z}_X)$, and let $(U_t)_{t \in \mathbf{R}}$ be a family of open subsets of X . We assume the following conditions.

- (i) $U_t = \bigcup_{s < t} U_s$, for all $t \in \mathbf{R}$.
- (ii) For all pairs (s, t) with $s \leq t$, the set $\overline{U_t - U_s} \cap \mathrm{supp} F$ is compact.
- (iii) Setting $Z_s := \bigcap_{t > s} \overline{U_t - U_s}$, we have for all pairs (s, t) with $s \leq t$, and all $x \in Z_s - U_t$,

$$(H_{X - U_t}^j(F))_x = 0 \quad \text{for all } j \leq k.$$

- (iv) For any $s \in \mathbf{R}$, $\bigcap_{t>s} U_t = \overline{U_s}$ and $\overline{U_s}$ is closed in a paracompact open subset of X .
(v) ^{*6} $Z_s - U_s \subset \partial U_s$ for all $s \in \mathbf{R}$.

Then, for all $t \in \mathbf{R}$, we have the isomorphisms

$$H^j \left(\bigcup_s U_s; F \right) \xrightarrow{\sim} H^j(U_t; F) \quad \text{for all } j < k,$$

and the injections

$$H^k \left(\bigcup_s U_s; F \right) \hookrightarrow H^k(U_t; F).$$

Proof. Consider the following conditions for $j \in \mathbf{Z}$, s , and t :

$$\begin{aligned} \mu_s^j &: \varinjlim_{t>s} H^j(U_t; F) \xrightarrow{\sim} H^j(U_s; F), & (a)_s^j \\ \mu_s^k &: \varinjlim_{t>s} H^k(U_t; F) \hookrightarrow H^k(U_s; F), & (a')_s^k \\ \lambda_s^j &: H^j(U_t; F) \xrightarrow{\sim} \varprojlim_{s<t} H^j(U_s; F). & (b)_s^j \end{aligned}$$

Let us prove $(a)_s^j$ for $s \in \mathbf{R}$ and $j < k$. We can assume that $\overline{U_t - U_s}$ is compact for any $s \leq t$ by replacing X by $\text{supp } F$. Consider the distinguished triangle

$$\text{R}\Gamma_{X-U_t}(F) \rightarrow F \rightarrow \text{R}\Gamma_{U_t}(F) \xrightarrow{+1}.$$

Applying to this triangle the functor $(\cdot)_{\overline{U_t}}$, we have

$$(\text{R}\Gamma_{X-U_t}(F))_{\overline{U_t}} \rightarrow F_{\overline{U_t}} \rightarrow \text{R}\Gamma_{U_t}(F) \xrightarrow{+1}.$$

Again applying to this triangle the functor $\text{R}\Gamma(X; \cdot)$, we have

$$\text{R}\Gamma(X; (\text{R}\Gamma_{X-U_t}(F))_{\overline{U_t}}) \rightarrow \text{R}\Gamma(\overline{U_t}; F) \rightarrow \text{R}\Gamma(U_t; F) \xrightarrow{+1}. \quad (5.1)$$

By hypotheses (iii)–(v), it follows that

$$\left(H_{X-U_t}^j(F) \right)_{\partial U_t} = 0$$

for any $j \leq k$ and then

$$H^j(X; (\text{R}\Gamma_{X-U_t}(F))_{\overline{U_t}}) = 0 \quad \text{for any } j \leq k.$$

^{*7} Thus for any t , we have

$$H^j(\overline{U_t}; F) \xrightarrow{\sim} H^j(U_t; F) \quad \text{for any } j < k, \quad (5.2)$$

$$H^k(\overline{U_t}; F) \hookrightarrow H^k(U_t; F). \quad (5.3)$$

^{*6} This condition might be deduced from (iii).

^{*7} この辺の議論はきちんと詰めないといけない。

Therefore, we have by (iv) and (5.2),

$$\begin{aligned}\varinjlim_{t>s} H^j(U_t; F) &\cong H^j\left(\bigcap_{t>s} U_t; F\right) \\ &= H^j(\overline{U_s}; F) \\ &\xrightarrow{\sim} H^j(U_s; F)\end{aligned}$$

for any $j < k$, and $(a)_s^j$ follows. Similarly, we have by (iv) and (5.3),

$$\begin{aligned}\varinjlim_{t>s} H^k(U_t; F) &\cong H^k\left(\bigcap_{t>s} U_t; F\right) \\ &= H^k(\overline{U_s}; F) \\ &\hookrightarrow H^k(U_s; F)\end{aligned}$$

and $(a')_s^k$ follows. □

Remark 5.2. Lemma 4.2 follows from Proposition 5.1. Indeed, putting $X =]-\infty, b[$, $F = G|_{I_b}$, and $U_t = I_t$, we find that

- (i) $\bigcup_{s < t} I_s = \bigcup_{s < t}]-\infty, s[=]-\infty, t[= I_t$ for all $t \in \mathbf{R}$.
- (ii) If $s \leq t$, then $\overline{I_t - I_s} \cap \text{supp } G = \overline{[s, t[} \cap \text{supp } G = [s, t] \cap \text{supp } G$ is compact.
- (iii) For $s \in \mathbf{R}$, we have

$$Z_s = \bigcap_{t > s} \overline{I_t - I_s} = \bigcap_{t > s} \overline{[s, t[} = \bigcap_{t > s} [s, t] = \{s\},$$

and

$$Z_s - I_t = \{s\} -]-\infty, t[= \begin{cases} \{s\} & \text{if } s = t, \\ \emptyset & \text{if } s < t. \end{cases}$$

By hypothesis of Lemma 4.2, for all $s \in \{s\}$,

$$H_{X - I_s}^j(F)_s = H_{[s, \infty[}^j(F)_s = 0.$$

Then we can apply Proposition 5.1, and we have the isomorphisms for all $t \in [a, b[$ and $j < k$,

$$H^j(I_b; F) = H^j\left(\bigcup_s I_s; F\right) \xrightarrow{\sim} H^j(I_t; F)$$

and the injections

$$H^k(I_b; F) = H^k\left(\bigcup_s I_s; F\right) \hookrightarrow H^k(I_t; F)$$

6 Relative Microlocal Morse Theory via Truncated Microsupport

Let $f: Y \rightarrow X$ be a morphism of manifolds, and φ a real valued function of class C^1 . For a real number $t \in \mathbf{R}$, we set

$$\begin{aligned} Y_t &:= \{y \in Y; \varphi(y) < t\} = \varphi^{-1}(]-\infty, t[), \\ \tilde{Y}_t &:= \{y \in Y; \varphi(y) \leq t\} = \varphi^{-1}(]-\infty, t]). \end{aligned}$$

Let us define the following morphisms:

$$\begin{array}{ccc} Y_t & \xrightarrow{j_t} & Y \\ & \searrow f_t & \downarrow f \\ & & X, \end{array} \quad \begin{array}{ccc} \tilde{Y}_t & \xrightarrow{\tilde{j}_t} & Y \\ & \searrow f_t & \downarrow f \\ & & X. \end{array}$$

Proposition 6.1. Let $G \in D^b(Y)$, and assume $f_t: \text{supp}(G) \cap \overline{Y}_t \rightarrow X$ is proper for all $t \in \mathbf{R}$. Let k be an integer and t_0 a real number.

- (i) Assume that, for any $y \in Y - \tilde{Y}_{t_0}$, we have

$$d\varphi(y) \notin \left(\text{SS}_k(G) + f_d \left(Y \times_X T^*X \right) \right).$$

Then

$$R^j f_* G \cong R^j f_{t_*} j_t^{-1} G \quad \text{for } j < k \text{ and } t > t_o, \quad (\text{a})$$

$$R^k f_* G \hookrightarrow R^k f_{t_*} j_t^{-1} G \quad \text{for } t > t_o, \quad (\text{a}_k)$$

$$R^j \tilde{f}_* G \cong R^j \tilde{f}_{t_*} \tilde{j}_t^{-1} G \quad \text{for } j < k \text{ and } t \geq t_o, \quad (\text{a}')$$

$$R^k \tilde{f}_* G \hookrightarrow R^k \tilde{f}_{t_*} \tilde{j}_t^{-1} G \quad \text{for } t \geq t_o, \quad (\text{a}'_k)$$

$$\begin{aligned} \text{SS}_k(Rf_* G) &\subset f_\pi \left(f_d^{-1} \left(\text{SS}_k(G) \cap \pi^{-1}(\tilde{Y}_{t_o}) \right) \right) \\ &= f_\pi f_d^{-1} (\text{SS}_k(G)). \end{aligned} \quad (\text{b})$$

(ii) Assume that, for any $y \in Y - \tilde{Y}_{t_o}$, we have

$$-d\varphi(y) \notin \left(\text{SS}_k(G) + f_d \left(Y \times_X T^* X \right) \right).$$

Then

$$R^j f_! G \cong R^j f_{t_!} j_t^{-1} G \quad \text{for } j < k \text{ and } t > t_o, \quad (\text{c})$$

$$R^k f_! G \hookrightarrow R^k f_{t_!} j_t^{-1} G \quad \text{for } t > t_o, \quad (\text{c}_k)$$

$$R^j \tilde{f}_! G \cong R^j \tilde{f}_{t_*} \tilde{j}_t^! G \quad \text{for } j < k \text{ and } t \geq t_o, \quad (\text{c}')$$

$$R^k \tilde{f}_! G \hookrightarrow R^k \tilde{f}_{t_*} \tilde{j}_t^! G \quad \text{for } t \geq t_o, \quad (\text{c}'_k)$$

$$\begin{aligned} \text{SS}_k(Rf_! G) &\subset f_\pi \left(f_d^{-1} \left(\text{SS}_k(G) \cap \pi^{-1}(\tilde{Y}_{t_o}) \right) \right) \\ &= f_\pi f_d^{-1} (\text{SS}_k(G)). \end{aligned} \quad (\text{d})$$

Proof. Let $\tilde{f}$ be the morphism $(f, \varphi): Y \rightarrow X \times \mathbf{R}$, and let q and $\tilde{\varphi}$ be the first and second projections defined on $X \times \mathbf{R}$ respectively. □

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7 はじめに

修士論文は指導教員の指導のもとに作成する。修士論文の構成や文体については指導教員の指導に従う。一般的にはセミナーで読んだ論文や書籍を参考にするとよい。数学に限らず、文章を書く際の注意として参考文献

献 [2] は定評のある和書である．このサンプルファイルは一般的な修士論文の構成を紹介するものである．

8 主結果

以下に修士論文作成に際しての注意および L^AT_EX を利用する際の注意点を示す．

8.1 修士論文作成の注意

実際の修士論文作成に当たっては指導教員とよく相談し，最初の草稿は短く作成し早めに見てもらおうとよい．短く作成し，早く見ってもらうことで論文調の文章作成に慣れることができ，短く作成することで構成を練ることができる．

論文では句読点にカンマとピリオドを用いる．半角・全角は統一する．

8.2 L^AT_EX 利用上の注意

段落分けの注意を示す．L^AT_EX で段落を分ける際には 1 行空ければよい．段落の冒頭には自動的に 1 文字分の空白が入る．本文中で強制的な改行指示 (\\) を使う場面はごく少ない．一般的に，1 つの段落には 1 つの内容を記述し，冒頭の文章で段落の内容を簡潔に紹介する．

節や数式，図表，参考文献の引用について説明する．節 (\section など) 番号，数式 (equation 環境) 番号，図表の番号は自動的に設定される．このサンプルファイルのように label をつけて適切に参照する．まず，節番号の参照 (第 7 節) と参考文献の参照 [1] を例として示す．

続いて，式 (8.1) を主結果の例として，数式番号の参照例を紹介する．

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} f(T^k(x)) = \int_X f d\mu \quad (8.1)$$

番号を振らない数式は次のように書く．

$$\lambda(x) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} \log |f'(T^k(x))|.$$

表を参照する場合の例 (表 1) を示す．図の場合も figure 環境を使えば同様である．

$a = 1$	$a = 2$
成立	不成立

表 1 表の例

定義 1. 定義は defn 環境を使う．

命題 2. 命題は prop 環境．

補題 3. 補題は lemma 環境．

主定理 4 を下記に述べる．

定理 4 (主定理). 命題，補題，定理は amsthm パッケージを使う．本ファイル冒頭の記述を参照．必要があれば newtheorem を用いて同じように増やして良い．

目次, 参考文献を正しく参照するために, platex の処理は 3 回繰り返すこと.

9 終わりに

修士論文の構成例を簡単に紹介し, L^AT_EX を使用する際の注意点を示した. L^AT_EX 全般の参考書としては [3] が有名である. PC へのインストール用 DVD も付属している.

謝辞

本修士論文を作成するにあたり, ご指導いただいた指導教員の OO 先生に深く感謝します.

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